

Report 2: Sentence Memorability Experiment

Idle Death Gamblers

Shreyas Deb Yajat Lakhanpal Prasoon Dev

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1 Introduction

1.1 Sentence Memory: Gist vs. Wording

Human memory for language is not a faithful archive of words and syntax. A central finding in cognitive psychology is that listeners and readers retain the *meaning* (gist) of sentences with much greater fidelity than their verbatim wording or grammatical structure. The *kernel-plus-tag* hypothesis (Miller, 1962; Mehler, 1963) proposed that sentences are encoded as a semantic kernel with syntactic features stored as forgettable tags. Bregman and Strasberg (1968) refuted this: when participants misidentified the syntactic form on a first guess, their second guesses exceeded chance ($\chi^2(4) = 20.8, p < .001$), suggesting that grammatical form is *reconstructed* from imagery and noun salience rather than retrieved as a tag.

Begg (1971) tested the gist–wording dissociation directly via a **continuous recognition paradigm**. Meaning judgments declined sharply with lag ($r = -.996$), while wording judgments were uncorrelated with lag ($r = -.107$), interpreted as evidence that concrete sentence meaning is stored as a spatial image and wording is reconstructed from that image trace at retrieval.

If wording is reconstructed from an imagistic trace, the imageability of constituent nouns becomes critical. James et al. (1973) tested this in free recall using four noun-imageability conditions (HH, HL, LH, LL) and documented a threshold-like recall pattern and a bias toward placing salient nouns in subject position during reconstruction.

1.2 Signal Detection Theory Motivation

In recognition-memory tasks, raw accuracy or corrected hit-minus-false-alarm scores can obscure an important distinction between two processes: how well participants discriminate old from new items, and how conservatively they choose to respond “old.” Signal Detection Theory (SDT; Green and Swets 1966) addresses this by separating **sensitivity** (d') from **response criterion** (c), and has long been the standard framework for recognition-memory research (Macmillan and Creelman, 2005). We therefore use SDT as a downstream analysis to test whether the observed noun-condition differences reflect a genuine change in discriminability (d') or instead arise from shifts in response bias (c). The non-parametric index A' (Snodgrass and Corwin, 1988) is included as a robustness measure that requires no Gaussian equal-variance assumption.

1.3 Core Question and Hypotheses

The present study extends James et al. (1973) from free recall to online continuous recognition, tests Begg (1971)’s prediction that imageability selectively affects gist but not wording, and uses SDT to decompose the imageability effect into sensitivity and criterion components.

H1: (Gist): Corrected IR differs across noun imageability conditions (HH, HL, LH, LL), with LL showing the largest deficit.

H2: (Wording): WR accuracy is not significantly affected by noun imageability; voice is reconstructed from the imagistic trace.

H3: (Voice): Active-voice sentences will show different memorability and wording-recognition patterns compared to passive.

H4: (No Interaction): Voice does not moderate the noun imageability effect on Corrected IR or WR.

2 Dataset and Experimental Design

2.1 Participants

Data were collected from 114 online participants. Two participants failed the vigilance criterion on all three blocks and were excluded, leaving **112 participants** (block pass rate: 96.2%, 329/342 blocks).

2.2 Design

Fully within-participants **4 (Noun Condition: HH, HL, LH, LL) $\times$ 2 (Voice: Active, Passive)** design. Noun conditions were defined by crossing subject and object noun imageability (Table 1). High-Frequency (HF) filler sentences served as lures for false alarm estimation.

Code	Subject Noun	Object Noun
HH	High imageability	High imageability
HL	High imageability	Low imageability
LH	Low imageability	High imageability
LL	Low imageability	Low imageability

Table 1: Four experimental noun conditions.

2.3 Stimuli

Simple SVO sentences were constructed by crossing noun imageability levels with voice. High- and low-imageability nouns were drawn from established word memorability norms; verbs were medium-concreteness action verbs. After fluency and coherence filters, 200 active-voice sentences (50 per condition) were selected and each converted to passive, yielding **400 target sentences** plus HF lure sentences.

2.4 Procedure

Participants completed a continuous recognition task across **3 experimental blocks** preceded by a practice block. Each sentence was displayed for ≈ 4000 ms (ITI: 700–1300 ms). For each probe, participants made two sequential judgments:

1. **IR:** “Have you seen this sentence before?” (spacebar).
2. **WR:** “Same grammatical voice?” (key press; elicited only when IR was pressed).

2.5 Validation and Exclusions

Raw event logs (81,329 rows) were processed in R. Block boundaries were identified from “Rest Phase started” markers; per-trial accuracy variables were extracted from **IR pressed** and **WR pressed** rows. Each block was retained only if:

$$\text{Correct Validation IRs} > \frac{\text{Wrong Validation IRs}}{2} + \text{Missed Validation IRs}$$

Failed blocks were excluded; two participants failing all blocks were removed entirely, leaving 48,281 observations. The design was balanced: each noun condition contributed 1,316 encoding trials; each voice condition 2,632 trials.

Practice block exclusion. Practice blocks were excluded from all inferential analyses because they were designed only to familiarise participants with the response interface, not to measure stable sentence-memory performance. Including them would introduce variance due to interface acclimatisation (learning key assignments, calibrating response pacing) that is unrelated to noun imageability processing. The block-learning analysis tests changes across the three *experimental* blocks; this question is only answerable cleanly if every observation reflects fully-familiarised participants responding to proper experimental stimuli.

3 Methods

3.1 Scoring

False alarm counts were computed from IR **pressed** events on HF first-showing trials:

$$\widehat{\text{FA rate}} = \frac{\text{IR presses on HF first-showings}}{\text{HF first-showing display events}} \quad (M = 0.160, SD = 0.104, \text{ range: } 0.00\text{--}0.43)$$

Three dependent variables were derived per participant per cell:

1. **Corrected IR:** $\text{IR}_{CR} = \text{Hit Rate} - \text{FA Rate}$.
2. **WR Accuracy:** Proportion of correct voice judgments, conditional on IR pressed.
3. **IR Reaction Time (RT):** Mean RT on correct target probes.

3.2 Primary Non-Parametric Analysis

Shapiro-Wilk tests per condition group confirmed that Corrected IR and WR Accuracy are non-normal in all four groups ($p < .001$), expected given that these are bounded proportions computed from 3–6 trials per cell. Non-normality justifies **Friedman’s test** as the omnibus statistic. Friedman’s is the within-participants equivalent of Kruskal-Wallis: it blocks on participant identity, ranks observations within each participant, and tests for systematic rank differences across conditions without distributional assumptions.

Three omnibus Friedman tests (IR CR, WR Accuracy, IR RT) require a Bonferroni correction: $\alpha_{FW} = 0.05/3 = .0167$. Within significant omnibus tests, Holm-corrected paired Wilcoxon signed-rank post-hoc tests were conducted. Noun condition $\times$ voice interactions were evaluated with **Scheirer-Ray-Hare (SRH) tests** (the non-parametric two-way analogue of ANOVA; cell balance confirmed, $N = 112$ per cell). SRH was designed for independent groups; sensitivity Friedman tests per voice level follow as a robustness check.

3.3 SDT Analysis

Rationale. The corrected IR score is useful as a bias-reduced behavioural summary but does not fully separate memory strength from response strategy. SDT provides that decomposition by estimating sensitivity (d') and criterion (c) from hit and false-alarm rates (Stanislaw and Todorov, 1999; Macmillan and Creelman, 2005). This allows us to test whether the LL deficit reflects weaker memory evidence, a more conservative response threshold, or both.

SDT parameters were computed per participant per noun condition using log-linear correction (adding 0.5 to counts, 1 to denominators) to prevent infinite z -values (Macmillan and Creelman, 2005). The non-parametric A' index (Snodgrass and Corwin, 1988):

$$A' = 0.5 + \frac{(\text{HR} - \text{FAR})(1 + \text{HR} - \text{FAR})}{4 \cdot \text{HR} \cdot (1 - \text{FAR})} \quad (\text{when } \text{HR} \geq \text{FAR})$$

is bounded in $[0.5, 1]$ and requires no equal-variance assumption, serving as an independent check on d' conclusions. Because d' and c passed normality per group ($p > .09$), **repeated-measures ANOVA** was used for those indices; A' uses Friedman's. Convergent validity was checked via Spearman $\rho(d', \text{Corrected IR})$.

Caveat: All HF lures are active-voice; a global FA rate is therefore used for both voice conditions, which may marginally inflate d' for passive sentences. Voice SDT comparisons are interpreted with this limitation in mind.

3.4 GLM Analysis

GLMs were fit at cell level (one row per participant $\times$ noun condition $\times$ voice; $N = 890$ after listwise exclusion of 6 rows with no WR trials). Reference levels: noun condition = LL, voice = Active. Backward AIC stepwise elimination from a full model including main effects and the noun condition $\times$ voice interaction. Breusch-Pagan tests indicated heteroscedasticity for both outcomes (IR: $p = .040$; WR: $p = .002$); **HC3 robust standard errors** were applied throughout. No Cook's D exceeded 1. Bayesian model comparison used `lmBF()` from the `BayesFactor` package.

3.5 Block-Learning Analysis

The block-learning analysis tests whether gist recognition improves across the three experimental blocks and whether the LL deficit is stable or attenuates with repeated exposure. Three tests were applied: (a) Friedman's on block main effect (collapsed across noun condition); (b) per-block Friedman on noun condition; (c) SRH condition $\times$ block interaction.

4 Results

4.1 Descriptive Statistics

Data from **112 participants**, 329 validated blocks (96.2%), and **896 participant-by-cell observations**. Global FA rate: $M = 0.160$ ($SD = 0.104$). All condition groups were non-normal (Shapiro-Wilk $p < .001$), confirming non-parametric inference.

Cond	M (IR CR)	SD	Mdn	M (WR Acc)	SD	Mdn
HH	0.689	0.210	0.714	0.747	0.208	0.772
HL	0.691	0.200	0.716	0.747	0.209	0.792
LH	0.679	0.209	0.703	0.762	0.203	0.800
LL	0.632	0.224	0.638	0.730	0.220	0.757

Table 2: Descriptive statistics for Corrected IR and WR Accuracy, collapsed across voice.

4.2 Effect of Noun Condition on Corrected IR (H1)

Friedman's test revealed a significant main effect:

$$\chi^2(3) = 22.39, \quad p < .001, \quad \text{Kendall's } W = 0.067 \quad (\alpha_{FW} = .0167)$$

Post-hoc Holm-corrected paired Wilcoxon tests (Table 3) show a **threshold pattern**: LL is significantly below all other conditions; HH, HL, and LH are statistically indistinguishable (all Holm- $p \geq .446$). The presence of a single high-imageability noun is sufficient to anchor gist memory; a second provides no further benefit.

4.3 Effect of Noun Condition on WR Accuracy (H2)

Friedman's test: $\chi^2(3) = 3.34$, $p = .343$, $W = 0.010$ — non-significant at $\alpha_{FW} = .0167$. WR performance was robustly above chance in both voices (one-sample Wilcoxon vs. 0.50: $V > 6200$,

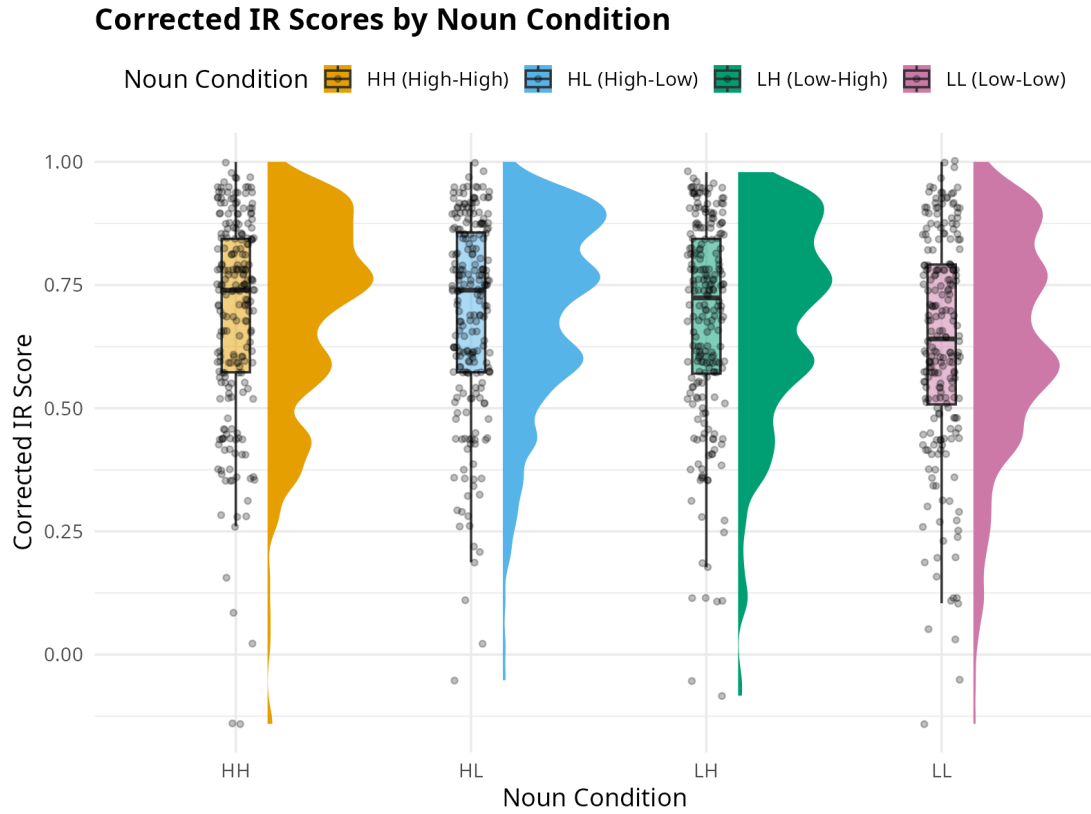


Figure 1: Corrected IR across the four noun conditions (raincloud plot). LL shows a visible deficit relative to HH, HL, and LH.

Pair	Holm p	Rank-biserial r	Magnitude
HH vs HL	1.000	.038	small
HH vs LH	1.000	.076	small
HH vs LL	.002	.360	moderate
HL vs LH	.446	.188	small
HL vs LL	<.001	.409	moderate
LH vs LL	.019	.268	small

Table 3: Post-hoc paired Wilcoxon tests on Corrected IR (Holm-corrected).

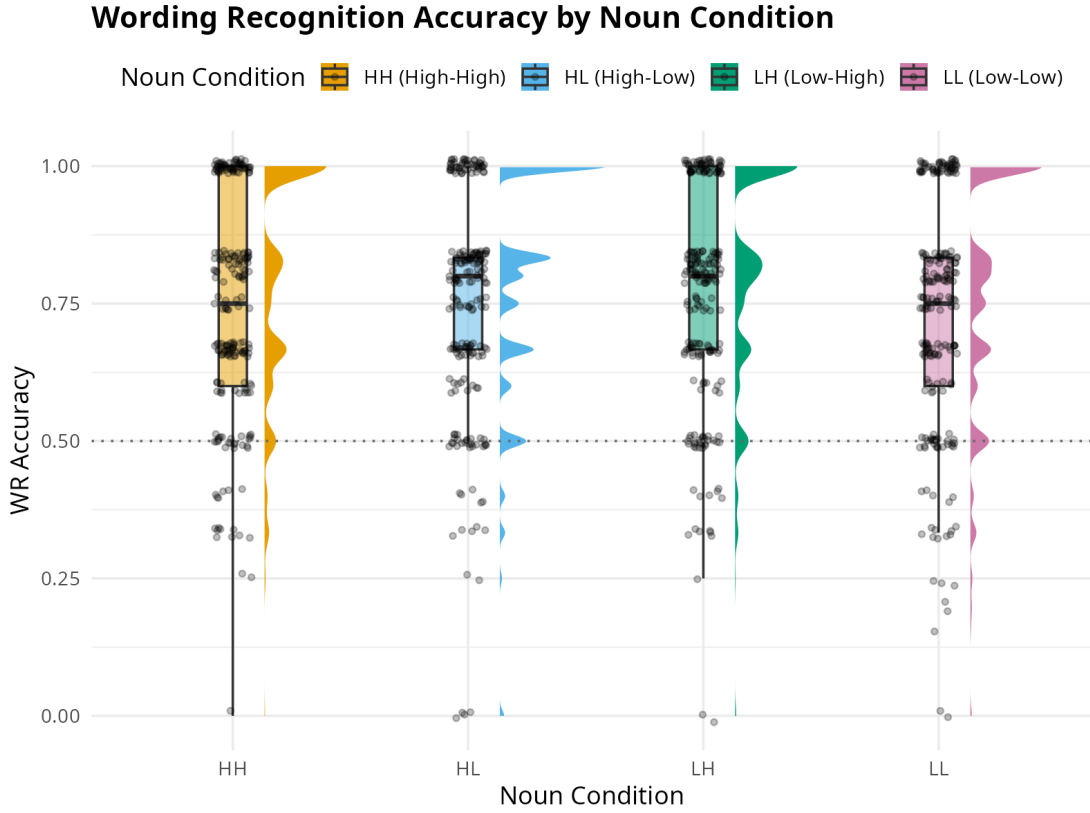


Figure 2: WR Accuracy across noun conditions. Performance is uniformly high and statistically indistinguishable across all conditions.

$p < .001$, $r > .83$). Noun imageability does not affect syntactic retention.

4.4 Noun Condition $\times$ Voice Interaction (H3 & H4)

SRH tests found for **Corrected IR**: noun condition main effect ($H = 11.26$, $p = .010$, $\eta_H^2 = 0.013$); no voice main effect ($H = 0.76$, $p = .384$); **no interaction** ($H = 0.22$, $p = .975$, $\eta_H^2 < .001$). For **WR accuracy**: no significant effects or interaction (all $p > .23$). These results support H4.

Paired Wilcoxon voice tests (collapsed across condition) found no significant voice effect on Corrected IR ($V = 1916.5$, $p = .485$, $r = .066$) or WR ($V = 3520.5$, $p = .114$, $r = .149$).

4.5 IR Reaction Time

Friedman's test on IR RT: $\chi^2(3) = 16.53$, $p < .001$, $W = 0.049$, significant at $\alpha_{FW} = .0167$. Post-hoc (Holm): LH vs. LL was the only significant pair ($p < .001$, $r = .387$). Passive LL sentences were slowest ($M = 1808$ ms vs. Active HL at 1664 ms), consistent with greater processing difficulty when both nouns are low-imageability.

4.6 SDT Results

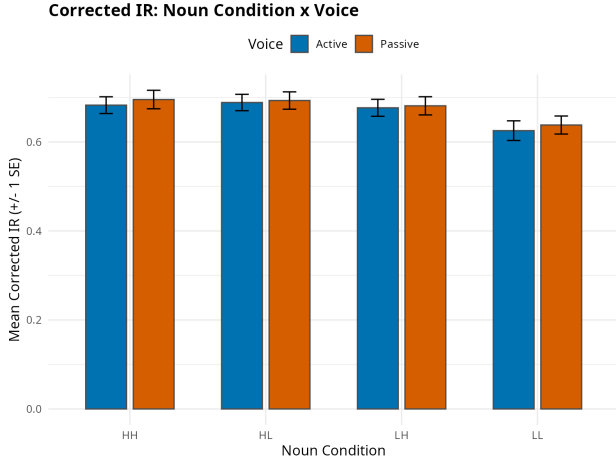
4.6.1 Descriptive SDT Statistics

4.6.2 Inferential Tests on d' and A'

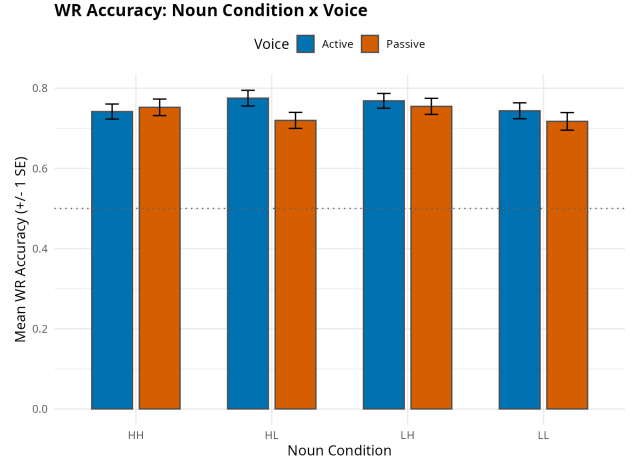
Shapiro-Wilk per group confirmed normality of d' ($p > .09$ all groups), justifying RM-ANOVA. A' is bounded and uses Friedman's.

RM-ANOVA on d' : Mauchly's $W = 0.961$, $p = .492$ (sphericity assumed).

$$F(3, 333) = 8.47, \quad p < .001, \quad \text{GES} = 0.018, \quad \text{Cohen's } f = 0.134 \text{ (small)}$$

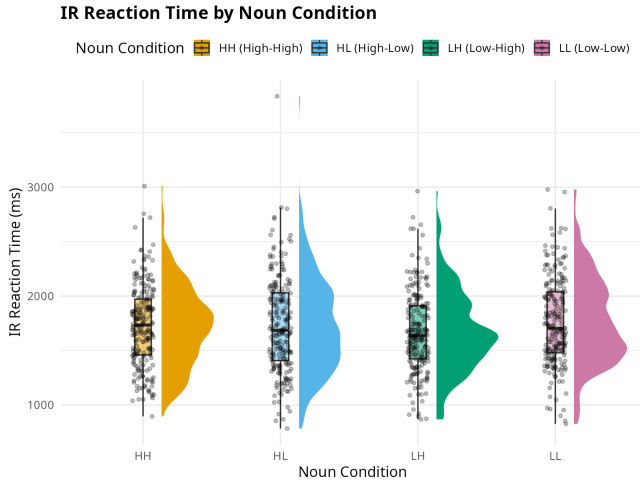


(a) Corrected IR by Condition $\times$ Voice

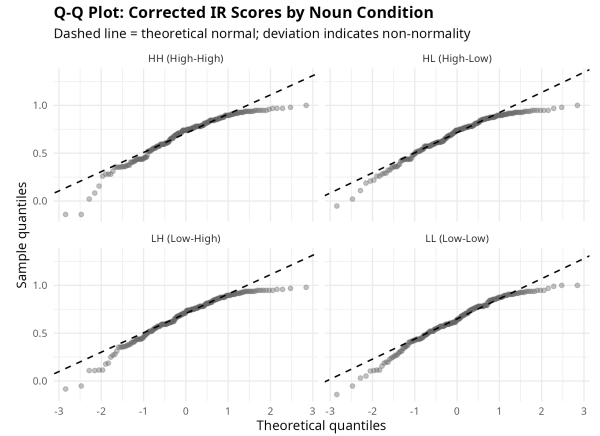


(b) WR Accuracy by Condition $\times$ Voice

Figure 3: Interaction plots. No systematic condition $\times$ voice moderation is visible for either outcome.



(a) IR RT by Condition



(b) Q-Q Plot: Corrected IR

Figure 4: RT distributions (left) and normality validation for Corrected IR (right).

Cond	d'_M	d'_{SD}	c_M	c_{SD}	A'_M	A'_{SD}
HH	2.146	0.690	-0.003	0.396	0.895	0.067
HL	2.146	0.654	-0.003	0.407	0.897	0.061
LH	2.091	0.698	0.024	0.393	0.891	0.068
LL	1.925	0.672	0.107	0.421	0.876	0.071

Table 4: SDT parameter estimates by noun condition ($N = 112$). d' = sensitivity; c = criterion ($c > 0$ = conservative); A' = non-parametric AUC (Snodgrass and Corwin, 1988).

Holm-corrected post-hoc paired t -tests: LL < HH ($p < .001$, $d = 0.39$), HL ($p < .001$, $d = 0.43$), LH ($p = .010$, $d = 0.29$); HH vs. HL ($p = .996$) and HH vs. LH ($p = .768$) non-significant. Threshold pattern confirmed.

Friedman’s on A' : $\chi^2(3) = 23.50$, $p < .001$, $W = 0.070$. Post-hoc Holm: LL < HH ($r = .360$), HL ($r = .415$), LH ($r = .256$); HH–HL, HH–LH non-significant ($p > .50$). The A' result, which makes no equal-variance assumption (Snodgrass and Corwin, 1988), replicates the d' pattern, confirming it is not an artefact of the SDT model’s distributional assumptions.

4.6.3 Criterion (c) Analysis

RM-ANOVA on c : $F(3, 333) = 8.47$, $p < .001$. The LL condition shows the most positive c ($M = 0.107$ vs. $c_{HH/HL} = -0.003$): participants required *more* subjective evidence before calling an LL sentence “seen.” This conservative shift occurs above and beyond the sensitivity reduction (d' is also lowest for LL), suggesting the LL deficit comprises both a weaker imaginal trace *and* a compensatory metacognitive adjustment in decision threshold.

4.6.4 Convergent Validity

Condition	ρ
HH	0.981
HL	0.976
LH	0.977
LL	0.960
Overall	0.977

Table 5: Spearman ρ between d' and Corrected IR ($N = 112$; all $p < .001$). $\rho \geq .96$ confirms Corrected IR as an adequate proxy for d' .

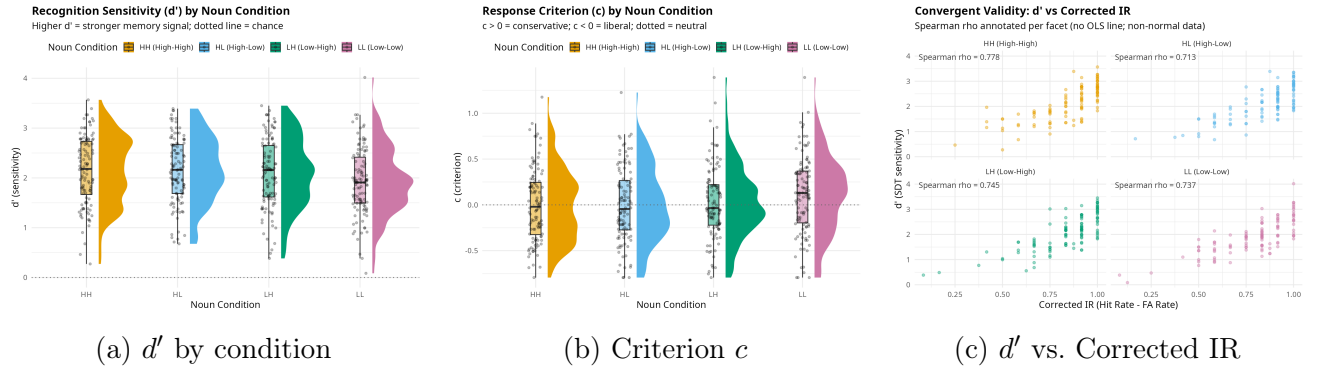


Figure 5: SDT results. (a) Sensitivity lower for LL. (b) Criterion shifts conservatively for LL. (c) Near-perfect $\rho = 0.977$ between d' and Corrected IR.

4.7 GLM Results

4.7.1 Corrected IR: Best Model

Backward AIC retained: Corrected IR $\sim$ noun_condition + mean_trial_position

$$F(4, 885) = 7.89, \quad p < .001, \quad R^2 = 0.034, \quad \text{Adj.}R^2 = 0.030$$

The near-equal $\hat{\beta}$ values for LH (0.049), HL (0.053), and HH (0.055) confirm the threshold structure in a covariate-controlled regression framework. Bayesian comparison: $\text{BF}_{10} = 10.76$ — positive evidence for noun condition predicting Corrected IR.

Term	$\hat{\beta}$	Robust SE	t	p	95% CI
Intercept	0.374	0.070	5.37	< .001	[0.237, 0.512]
Cond: LH	0.049	0.019	2.54	.011	[0.011, 0.088]
Cond: HL	0.053	0.020	2.73	.006	[0.015, 0.092]
Cond: HH	0.055	0.020	2.80	.005	[0.016, 0.094]
Trial position	0.011	0.003	3.88	< .001	[0.005, 0.017]

Table 6: HC3 coefficient estimates for best Corrected IR GLM. Coefficients for LH, HL, HH are nearly identical (threshold pattern). Bayesian model comparison: $BF_{10} = 10.76$ (positive evidence).

4.7.2 WR Accuracy: Best Model

Backward AIC retained: WR Accuracy $\sim$ voice + mean_trial_position

$$F(2, 887) = 4.24, \quad p = .015, \quad R^2 = 0.010$$

Noun condition was dropped by AIC. Bayesian comparison for noun condition: $BF_{10} = 0.001$ ($BF_{01} = 1027$) — very strong evidence noun condition does not predict WR.

4.7.3 Interaction Tests (H_4)

Wald F -tests (HC3) confirmed no noun condition $\times$ voice interaction:

- Corrected IR: $F(3, 880) = 0.098, p = .961; BF_{01} = 97$
- WR Accuracy: $F(3, 880) = 0.880, p = .451; BF_{01} = 35$

4.7.4 GLM Diagnostics

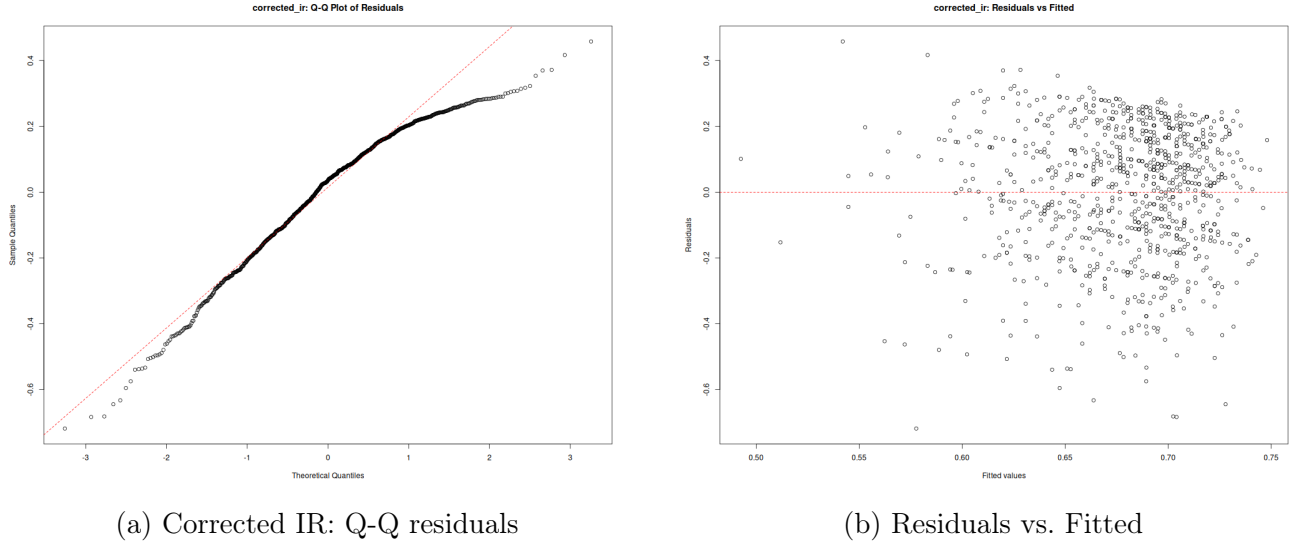


Figure 6: GLM diagnostic plots. HC3 robust SEs address heteroscedasticity; OLS point estimates remain unbiased. No Cook’s D exceeded 1; all VIF < 5.

4.8 Block-Learning Results (Control Analysis)

4.8.1 Block Main Effect

Friedman’s test (collapsed across noun condition) on Corrected IR:

$$\chi^2(2) = 23.30, \quad p < .001, \quad W = 0.111$$

Post-hoc Holm Wilcoxon: Block 1 < Block 2 ($p < .001$, $r = .351$) and Block 3 ($p < .001$, $r = .449$). Block 2 vs. 3 non-significant ($p = .166$, $r = .142$) — learning is front-loaded and plateaus after the first experimental block. No block effect on WR Accuracy ($\chi^2(2) = 1.42$, $p = .492$) or RT ($\chi^2(2) = 0.51$, $p = .773$).

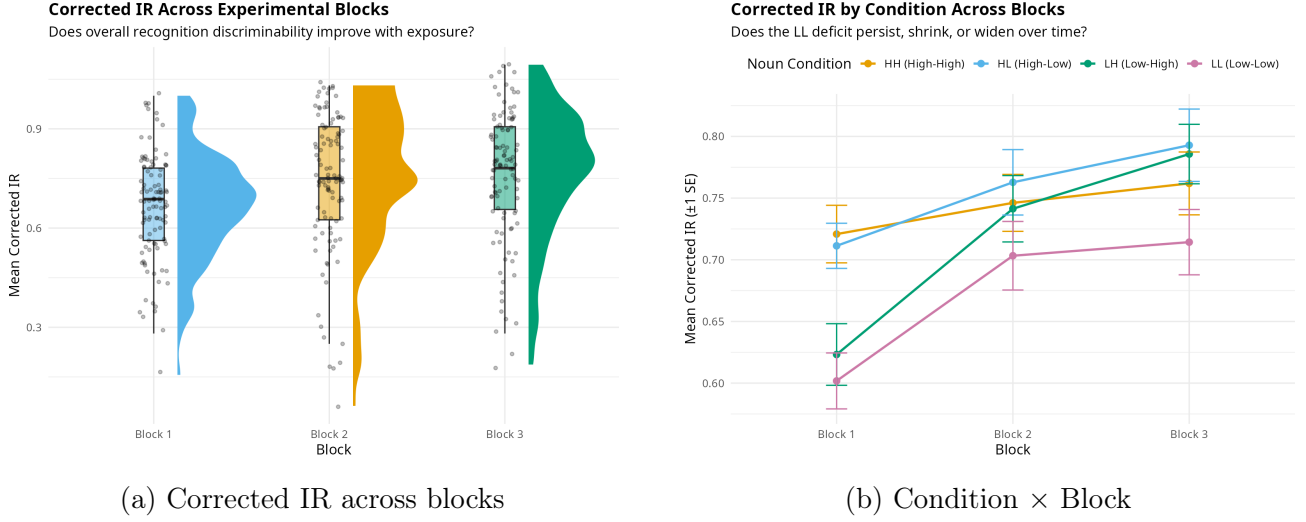


Figure 7: Block learning curves. IR improves Block 1 → 2 and plateaus. The noun condition ordering is stable across blocks; no condition × block interaction.

4.8.2 Noun Condition Effect Within Each Block

Block	$\chi^2(3)$	p	Kendall's W	Significant (α_{FW})?
Block 1	29.0	< .001	0.092	Yes
Block 2	8.06	.045	0.026	No ($p > .0167$)*
Block 3	11.0	.012	0.035	Yes

Table 7: Per-block Friedman tests for noun condition effect on Corrected IR ($N = 105$, complete data across all three blocks). *Block 2 is a descriptive trend only at the family-wise threshold.

4.8.3 Condition × Block Interaction

SRH: noun condition $H = 17.78$, $p < .001$; block $H = 51.39$, $p < .001$; **interaction** $H = 6.43$, $p = .377$. The LL deficit does not attenuate with repeated exposure — it is a structural property of the stimuli, not an artefact of early-block unfamiliarity.

5 Discussion and Conclusion

5.1 Summary of Findings

This study embedded the classic paradigms of Begg (1971) and James et al. (1973) into a large-scale online continuous recognition task.

The **threshold model** of noun imageability is confirmed across all four analytic frameworks. Corrected IR was impaired only when both subject and object nouns were low-imageability (LL; Friedman's $\chi^2(3) = 22.39$, $p < .001$). HH, HL, and LH were statistically indistinguishable (all Holm- $p \geq .446$). This replicated in d' (RM-ANOVA $F(3, 333) = 8.47$, $p < .001$) and A' (Friedman's $\chi^2(3) = 23.50$, $p < .001$), was stable across all three blocks (SRH interaction $p = .377$), and appeared in near-equal GLM coefficients ($\hat{\beta} \approx 0.05$ for all three non-LL conditions; $BF_{10} = 10.76$).

The SDT analysis adds a criterion component: LL sentences induce a **conservative response shift** ($c_{LL} = 0.107$) above and beyond reduced sensitivity. Participants appear aware of their weaker memory and require stronger subjective evidence before reporting recognition — a metacognitive adjustment invisible to Corrected IR alone.

Noun imageability had **no effect on WR accuracy**. WR was robustly above chance across all conditions (GLM: $BF_{01} = 1027$ for noun condition null; Friedman’s $p = .343$). Voice also had no effect on either outcome, and no condition $\times$ voice interaction was detected by any test.

5.2 Theoretical Implications

The meaning-wording dissociation replicates Begg (1971)’s central finding in an online paradigm. Meaning recognition is driven by the availability of an imagistic trace, anchored by at least one high-imageability noun; wording recognition is independent of trace strength, consistent with Begg’s proposal that exact wording is reconstructed at retrieval rather than stored. The absence of a voice effect contradicts the kernel-plus-tag hypothesis of Bregman and Strasberg (1968): active and passive sentences are equally recognised and equally discriminated, supporting the reconstruction account.

The threshold in recognition (vs. the gradient in James et al. (1973)’s free recall) likely reflects a task difference: free recall requires generating the sentence from memory (where every cue helps additively), while recognition requires only that a trace reaches decision threshold (where a single vivid anchor suffices).

5.3 Limitations and Future Directions

- **Sample size:** $N = 112$ is below the pre-registered target of 334; the study is underpowered for small WR and voice effects.
- **One-sided lures:** All HF lures are active-voice; passive d' may be marginally inflated.
- **SRH approximation:** SRH was designed for independent groups; its use in a within-participants design is an approximation.
- **Future:** Reach target N ; include passive-voice lures; apply Beta regression or trial-level logistic mixed-effects models; explore EEG correlates (N400) of the imageability effect.

Team Contributions

- **Shreyas Deb:** End-to-end R pipeline, log parsing, FA rate correction, score computation, visualisations, block learning analysis.
- **Yajat Lakanpal:** GLM pipeline (AIC selection, HC3 SEs, Bayesian model comparison), core non-parametric framework (Friedman, Wilcoxon, SRH, effect sizes).
- **Prasoon Dev:** Literature review and hypotheses, SDT analysis (d' , A' , c , convergent validity), Discussion and Theoretical Implications.

Code and Data Availability

Full pipeline and source: <https://github.com/shrey715/sentence-memorability>

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